2.12 Random numbers

random

The random module, in the Python Standard Library, provides met
values.

random()

The random() method returns a random floating-point value each
in the range 0 (inclusive) to 1 (exclusive).

Figure 2.12.1: Outputting two random floating-point numbers.

```
# imports random module
import random

# Generates a random floating point number
print(random.random())
print(random.random())
```

```
0.04675854711720506
0.944076660354665
```

randrange()

Python's randrange() method generates random integers within a

PARTICIPATION ACTIVITY 2.12.2: Restricting random integers to a specific number of values.

Start 2x speed

```
import random

# Generates random integers with 3 possible values
```

```
print(random.randrange(3))
print(random.randrange(3))
print(random.randrange(3))
print(random.randrange(3))
print(random.randrange(3))
```

```
2
1
0
0
2
```

Captions ^

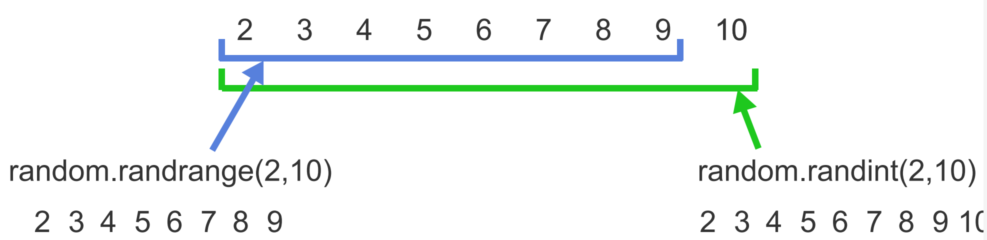
1. Each random.randrange(3) statement returns a random integer with 3 possible values: 0, 1,

Table 2.12.1: Random number modules for a defined range.

Function	C
randint(min, max) returns a random integer between min and max inclusive.	`# Returns a random integer` `print(random.randint(12, 20`
randrange(min, max) returns a random integer between min and max - 1 inclusive.	`# Returns a random integer` `print(random.randrange(12,`

PARTICIPATION ACTIVITY | 2.12.4: Generating random integers in a defined range not starting from 0.

Start 2x speed



Captions

1. A programmer wants to generate a random integer in the range 2 to 10 inclusive.
2. random.randrange(2,10) is only capable of generating 8 values in the range from 2 to 9.
3. The expression random.randint(2,10) is capable of generating all 9 values in the range from

Figure 2.12.2: Randomly moving a student from one seat to another.

```
#imports random module
import random
"""
Switch a student from a random seat on the left  (cols 1 to 15)
to a random seat on the right (cols 16 to 30)
Seat rows are 1 to 20
"""
rowNumL = random.randint(1, 20)  # 1 to 20 rows left
colNumL = random.randint(1, 15)  # 1 to 15  columns left
rowNumR = random.randint(1, 20)  # 1 to 20 rows right
colNumR = random.randint(16, 30) # 16 to 30 columns right

print(f"Move from row {rowNumL} col {colNumL} to row {rowNumR} col {colNumR}")
```

```
Move from row 18 col 13 to row 13 col 16
```

seed

For the first call to any random method, no previous random numb uses a built-in integer based on the current time, called a seed, to l number.

Figure 2.12.3: Using the same seed for each program run.

```
# import random module
import random

# Generate same sequence of numbers using same seed
random.seed(15)

# Generate same sequence of random integers between 1 and 10
print (random.randint(1, 10))
print (random.randint(1, 10))
print (random.randint(1, 10))
```

```
4
1
9

(next run)

4
1
9
```